

4 Fair Division

4.5 Sealed Bids Method (Cont.)

Example 4.5.1 Four small graphic design companies are merging operations to become one larger corporation. In this merger, there are a number of issues that need to be settled. Each company is asked to place a monetary value (in thousands of dollars) on each issue:

	Super Designs	DesignByMe	LayoutPros	Graphix
Company name	\$5	\$3	\$3	\$6
Company location	\$8	\$9	\$7	\$6
CEO	\$10	\$5	\$6	\$7
Chair of the Board	\$7	\$6	\$6	\$8

Example 4.5.2 Fair division does not always have to be used for items of value. It can also be used to divide undesirable items. Suppose Chelsea and Mariah are sharing an apartment, and need to split the chores for the household. They list the chores, assigning a negative dollar value to each item; in other words, the amount they would pay for someone else to do the chore (a per week amount). We will assume, however, that they are committed to doing all the chores themselves and not hiring a maid. Find a fair division if the bids are given as shown below.

	Chelsea	Mariah
Vacuuming	-\$10	-\$8
Cleaning bathroom	-\$14	-\$20
Doing Dishes	-\$4	-\$6
Dusting	-\$6	-\$4

Example 4.5.3 Four heirs (A, B, C, and D) must fairly divide an estate consisting of two items - a desk and a vanity - using the method of sealed bids. The players' bids (in dollars) are:

	A	B	C	D
Desk	320	240	300	260
Vanity	220	140	200	180

Example 4.5.4 After living together for a year, Emily, Kayla, and Kendra have decided to go their separate ways. They have several items they bought together to divide, as well as some moving-out chores. The values each place are shown below. Find the final allocation.

	Emily	Kayla	Kendra
Dishes	20	30	40
Vacuum cleaner	100	120	80
Dining table	100	80	130
Detail cleaning	-70	-40	-50
Cleaning carpets	-30	-60	-50